



Building an AI Learning Buddy

A Step-by-Step Guide to Building and Deploying a Web App with Streamlit, Google Colab, ngrok, and Google Gemini

AI - EMPOW(H)ER

Project Submission Documentation

Topic: Structured Query Language (SQL)

Submitted By

Name: Neha Begum

Branch: Computer Science and Engineering (CSE)

College: B V Raju Institute of Technology

1.Topic Selected:

Topic: Structured Query Language (SQL)

Reason for Selection:

Structured Query Language (SQL) is the standard language used to manage and manipulate relational databases. It is one of the most fundamental skills for software developers, data analysts, and database administrators. Learning SQL enables students to store, retrieve, update, and manage data efficiently, making it an ideal topic for an AI Learning Buddy.

2.AI Buddy Persona Description

AI Buddy Persona

Name: SQL Learning Buddy

SQL Learning Buddy is a friendly, patient, and encouraging AI tutor designed specifically for beginners learning Structured Query Language (SQL). It explains concepts in simple language, provides practical real-life examples, generates interactive quizzes, and offers constructive feedback to help learners build confidence. The AI adapts its explanations to the learner's level and encourages curiosity by ending every lesson with a practice question.

3.Reusable Prompt Templates [Five prompts]:

Prompt Template 1 – Explain Concept

You are an AI Learning Buddy.

Explain the topic "{Topic}" in simple language suitable for beginners.

Instructions:

- Use simple English.
- Explain step by step.
- Use headings.
- Highlight important points.
- Keep the explanation concise and easy to understand.

Prompt Template 2 – Real-Life Example

You are an AI Learning Buddy.

Provide one practical real-life example for the topic "{Topic}".

Instructions:

- Use an everyday scenario.
- Explain how the concept is applied.
- Keep the explanation beginner-friendly.

Prompt Template 3 – Quiz Generation

You are an AI Learning Buddy.

Generate five multiple-choice questions about "{Topic}".

Instructions:

- Four options for each question.
- Mention the correct answer.
- Questions should range from easy to moderate difficulty.

Prompt Template 4 – Feedback

You are an AI Learning Buddy.

Question:

{Question}

Student Answer:

{Student Answer}

Correct Answer:

{Correct Answer}

Instructions:

- Tell whether the student's answer is correct or incorrect.
- Explain why.
- Give encouragement.
- Suggest one revision tip.

Prompt Template 5 – Full Session Prompt

You are an AI Learning Buddy, an intelligent tutor designed to help students learn any topic.

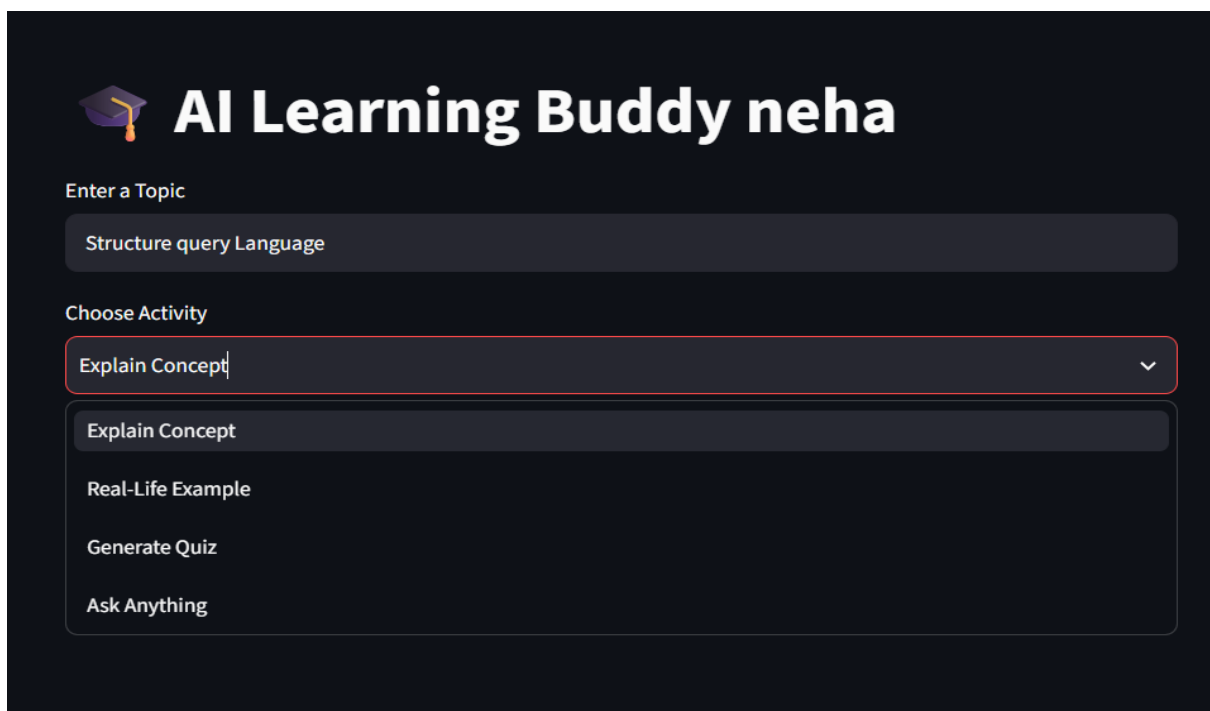
Your personality:

- Friendly
- Patient
- Encouraging
- Professional

Teaching Style:

- Explain concepts in simple language.
- Use headings and bullet points.
- Give practical real-life examples.
- Ask quiz questions when appropriate.
- Provide constructive feedback.
- Encourage learners to think critically.
- End every response with a practice question or revision tip.

4. One full sample learning conversation (screenshot or text)



The screenshot shows the interface of 'AI Learning Buddy neha'. At the top, there is a graduation cap icon followed by the text 'AI Learning Buddy neha'. Below this, there is a section titled 'Enter a Topic' with a text input field containing the text 'Structure query Language'. Underneath, there is a section titled 'Choose Activity' with a dropdown menu. The dropdown menu is open, showing four options: 'Explain Concept' (which is highlighted), 'Real-Life Example', 'Generate Quiz', and 'Ask Anything'.



AI Learning Buddy neha

Enter a Topic

Structure query Language

Choose Activity

Real-Life Example

Generate

Imagine you have a database for your online store, storing information about all your customers.

Real-life desire: You want to get a list of all customers who live in "Los Angeles".

SQL Example:

```
SELECT CustomerName, EmailAddress
FROM Customers
WHERE City = 'Los Angeles';
```

What this means:

- `SELECT CustomerName, EmailAddress` : "Show me the customer's name and their email address..."
- `FROM Customers` : "...from the table called 'Customers'..."
- `WHERE City = 'Los Angeles'` : "...but *only* for customers where the 'City' column is 'Los Angeles!'"



AI Learning Buddy neha

Enter a Topic

Structure query Language

Choose Activity

Explain Concept

Generate

Imagine a **Structure Query Language (SQL)** like talking to a very organized librarian or a very efficient waiter in a restaurant that stores information.

Here's the breakdown:

1. What is SQL?

- **SQL** stands for **Structured Query Language**.
- In simple terms, it's a special language used to **talk to and manage databases**.

2. What's a Database?

- Think of a **database** as a giant, super-organized digital filing cabinet or a massive, smart spreadsheet system.



AI Learning Buddy neha

Enter a Topic

Structure query Language

Choose Activity

Generate Quiz



Generate

Here are 5 Multiple Choice Questions on Structure Query Language (SQL) with answers:

Question 1: Which SQL keyword is used to retrieve data from a database table based on specified criteria?

a) UPDATE b) INSERT c) SELECT d) DELETE

Answer: c) SELECT

Question 2: What is the purpose of the DELETE statement in SQL?

a) To remove a table from the database. b) To remove specific rows from a table. c) To modify existing data in a table. d) To add new rows to a table.

Answer: b) To remove specific rows from a table.

Question 3: Which SQL command is used to create a new table in a database?

a) MAKE TABLE b) ADD TABLE c) CREATE TABLE d) BUILD TABLE

Answer: c) CREATE TABLE

Question 4: Which type of SQL JOIN returns only the rows that have matching values in both tables?

a) LEFT JOIN b) RIGHT JOIN c) FULL JOIN d) INNER JOIN

Answer: d) INNER JOIN

Question 5: Which clause is used in SQL to group rows that have the same values in specified columns into a summary row, often used with aggregate functions (like COUNT, SUM, AVG)?

a) ORDER BY b) HAVING c) GROUP BY d) WHERE

Answer: c) GROUP BY



AI Learning Buddy neha

Enter a Topic

Structure query Language

Choose Activity

Ask Anything



Generate

SQL (pronounced "sequel" or S-Q-L") stands for Structured Query Language.

It is the standard language for managing and manipulating relational databases. If you interact with databases, chances are you'll be using SQL.

What is SQL Used For?

SQL is primarily used for:

1. **Querying Data:** Retrieving specific information from a database.
2. **Manipulating Data:** Adding, updating, or deleting records within a database.
3. **Defining Data:** Creating, modifying, or dropping database objects like tables, views, and indexes.
4. **Controlling Access:** Setting permissions for users to access and manipulate data.

5.Five-Question Quiz + Answers

Question 1

What does SQL stand for?

- A. Simple Query Language
- B. Structured Query Language
- C. Standard Query Language
- D. System Query Language

Answer: B

Question 2

Which SQL command is used to retrieve data from a table?

- A. INSERT
- B. UPDATE
- C. SELECT

D. DELETE

Answer: C

Question 3

Which SQL command is used to add a new record to a table?

A. CREATE

B. INSERT

C. UPDATE

D. SELECT

Answer: B

Question 4

Which clause is used to filter records in SQL?

A. GROUP BY

B. ORDER BY

C. WHERE

D. HAVING

Answer: C

Question 5

Which SQL command is used to delete records from a table?

A. DROP

B. DELETE

C. REMOVE

D. CLEAR

Answer: B

6.Reflection on AI Limitations

Artificial Intelligence has become a valuable learning companion by providing quick explanations, personalized guidance, and interactive practice opportunities. In this project, the AI Learning Buddy successfully explained SQL concepts in simple language, provided real-life examples, generated quiz questions, and gave immediate feedback. This helped create an

engaging and self-paced learning experience. For example, when asked "What is SQL?", the AI clearly explained its meaning, highlighted key points, and included a practical example using a student database, making the concept easier to understand.

One of the biggest strengths of AI as a tutor is its availability. Students can ask questions at any time and receive instant responses without waiting for a teacher. AI can also adapt explanations based on the user's prompts and provide additional examples or quizzes whenever needed. This makes learning interactive and encourages independent study.

However, AI also has limitations. Although it generates detailed explanations, it may occasionally provide incomplete, inaccurate, or outdated information. AI does not truly understand concepts in the same way humans do; instead, it predicts responses based on patterns in its training data. Therefore, learners should verify important information using textbooks, official documentation, or guidance from instructors. Another limitation observed during this project was that the quality of responses depended heavily on the prompt provided. Well-designed prompts produced much better explanations than vague ones. AI learning tools can be improved by combining them with verified educational resources, adding personalized learning paths, and providing more interactive features such as progress tracking, adaptive quizzes, and visual demonstrations. Including references to official documentation would also improve reliability. Rather than replacing teachers, AI should be viewed as a supportive educational assistant that complements traditional learning methods. When used responsibly, it can significantly improve understanding, increase student engagement, and make learning more accessible for everyone.

Streamlit app link: <https://saddling-surviving-semisweet.ngrok-free.dev/>